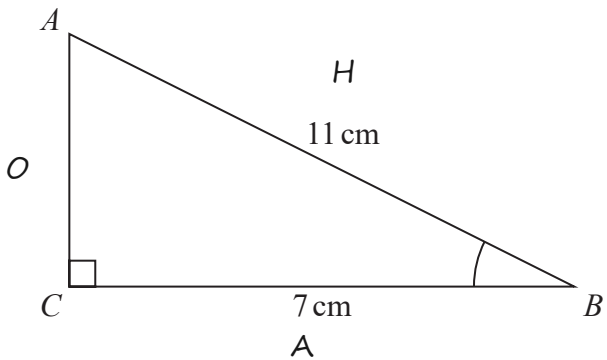
5 *ABC* is a right-angled triangle.



(a) Work out the size of angle *ABC*.
Give your answer correct to 1 decimal place.

Relative to angle B: adjacent = 7; hypotenuse = 11.

$\cos B = \frac{7}{11}$

1 mark
Use adjacent ÷ hypotenuse

$B = \cos^{-1}(7 \div 11) = 50.4788...°$

1 mark
50.5° to 1 decimal place

50.5 °

(2)

The length of the side *AB* is reduced by 1 cm.

The length of the side *BC* is still 7 cm.
Angle *ACB* is still 90°

(b) Will the value of cos *ABC* increase or decrease?
You must give a reason for your answer.

Increase: the new hypotenuse is 10 cm.

$\frac{7}{10} = 0.7$, which is greater than $\frac{7}{11} \approx 0.6364$.

(1)

1 mark
Compare the new and old cosine values

(Total for Question 5 is 3 marks)